

Pratham Arora

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Education

Plaksha University

B.Tech. in Computer Science & AI

Mohali, India — Expected Aug. 2026

CGPA: 8.19 / 10.0

Skills

Languages: Python, C++, SQL, JavaScript, TypeScript, C#, HTML/CSS, Bash

Web & Backend: FastAPI, Node.js, Next.js (SSR/CSR), REST APIs, GraphQL, Tailwind CSS

Cloud & Tools: Azure (Cosmos DB, OpenAI), Firebase, Git/GitHub, Linux, Vector Databases (FAISS, DiskANN), Vercel

AI/ML: PyTorch, Scikit-learn, LangChain, OpenCV, RAG Pipelines

Industry Experience

AI Intern

June 2025 – July 2025

Cotality | Kolkata

- Collaborated with one other intern to architect **FastAPI** backend parsing 50,000+ lines of code daily across 15+ programming languages via Abstract Syntax Tree (AST) analysis, enabling automated real-time documentation generation across polyglot codebases.
- Delivered 12 production **RESTful API endpoints** with response schemas enabling real-time code documentation preview in the client-facing web interface.
- Designed hash-based change detection system using **SHA-256 on AST nodes**, skipping unchanged functions – reducing AI embedding generation costs by **85%** and eliminating redundant vector database writes.
- Built **RAG pipeline** using LangChain, Azure OpenAI embeddings, and Cosmos DB with DiskANN vector search, migrating from FAISS as codebase scale grew.

Software Development Engineer Intern

June 2024 – July 2024

Orangewood Labs | Noida

- Built a modular LLM + robot control codebase for a food-preparation robotic arm, designing clean architectural boundaries that enabled parallel integration with a Computer Vision team working on real-time calibration.
- Integrated Computer Vision capabilities into the robotic control loop, optimizing calibration routines and improving arm positioning accuracy for complex food-preparation tasks.

Selected Technical Projects

AI Resume Builder & Interview Prep Tool | Next.js, Tailwind, Firebase, Gemini API

Jan. 2025 – Apr. 2025

- Built full-stack AI tool with real-time PDF preview and ATS-optimized resume generation, implementing **Firebase Authentication** with Google OAuth for secure user management across mobile and desktop.
- Integrated **Gemini 2.5 Flash** to power an “Interview GPT” feature that generates personalized interview questions by cross-referencing the user’s resume profile with a companion interview prep database (DSA problems, STAR templates) – implementing error handling for incomplete or ambiguous resume inputs.

Capacitated Vehicle Routing with Time Windows | Google OR-Tools, Google Maps API

Sept. 2023 – Dec. 2023

- Modelled a **multi-constraint VRP** end-to-end – formulating capacity, time-window, and depot constraints; tuning OR-Tools solver search strategies; and designing the full data pipeline from raw inputs to optimized routes.
- Integrated **Google Distance Matrix API** for real-world transit data and built interactive Maps visualizations for route validation and multi-stop delivery planning.

VR LLM Conversational Agent | Unity, Gemini API, Google Cloud STT/TTS

Aug. 2025 – Dec. 2025

- Engineered real-time VR conversational agent integrating **Gemini 2.5 Flash** with Google Cloud Speech-to-Text and Text-to-Speech, achieving average response latency of 1.8s through parallel API calls, caching, and audio preprocessing.
- Designed and ran **within-subjects user study** ($N = 18$) comparing three latency feedback mechanisms; co-authored paper submitted to MIT’s *Presence* journal (under review).

- Achieved **Top 100 ranking (83rd percentile)** in a DrivenData satellite imagery competition using **U-Net with EfficientNet-B3** backbone for semantic segmentation of kelp forests in noisy real-world satellite imagery.
- Built streak-detection pipeline using an ensemble of **CNN classifiers** (ResNet, DenseNet) with hard voting to filter corrupted training images, achieving **F1 score of 0.952**.

Leadership & Involvement

Head of Technology

Apr. 2025 – Present

Athleda – Plaksha Sports Society | Plaksha University

- Personally design and build a **sports platform serving 500+ students** alongside 2 junior developers – shipping event registration, photo galleries, and a live-scoring sports fest site with real-time score updates.
- Mentor junior developers through code reviews and pair programming, maintaining production quality across an actively growing platform.

LEAP.AI @ Plaksha

Feb. 2024 – Feb. 2025

Web Developer → Head of Web Development | Plaksha University

- Contributed as a core developer on a team of 7–8 building the AI club website, then led the team through **2 full redesigns** as Head, growing the platform to **800+ monthly users** for events, workshops, and project showcases.